Equipo y compilacion

lscpu

```
> wsl.exe -d Ubuntu-24.04 -- bash -lc "lscpu | head -n 15"
Architecture:                         x86_64
CPU op-mode(s):                       32-bit, 64-bit
Address sizes:                        48 bits physical, 48 bits virtual
Byte Order:                           Little Endian
CPU(s):                               12
On-line CPU(s) list:                  0-11
Vendor ID:                            AuthenticAMD
Model name:                           AMD Ryzen 5 4600H with Radeon Graphics
CPU family:                           23
Model:                                96
Thread(s) per core:                   2
Core(s) per socket:                   6
Socket(s):                            1
Stepping:                             1
BogoMIPS:                             5988.81

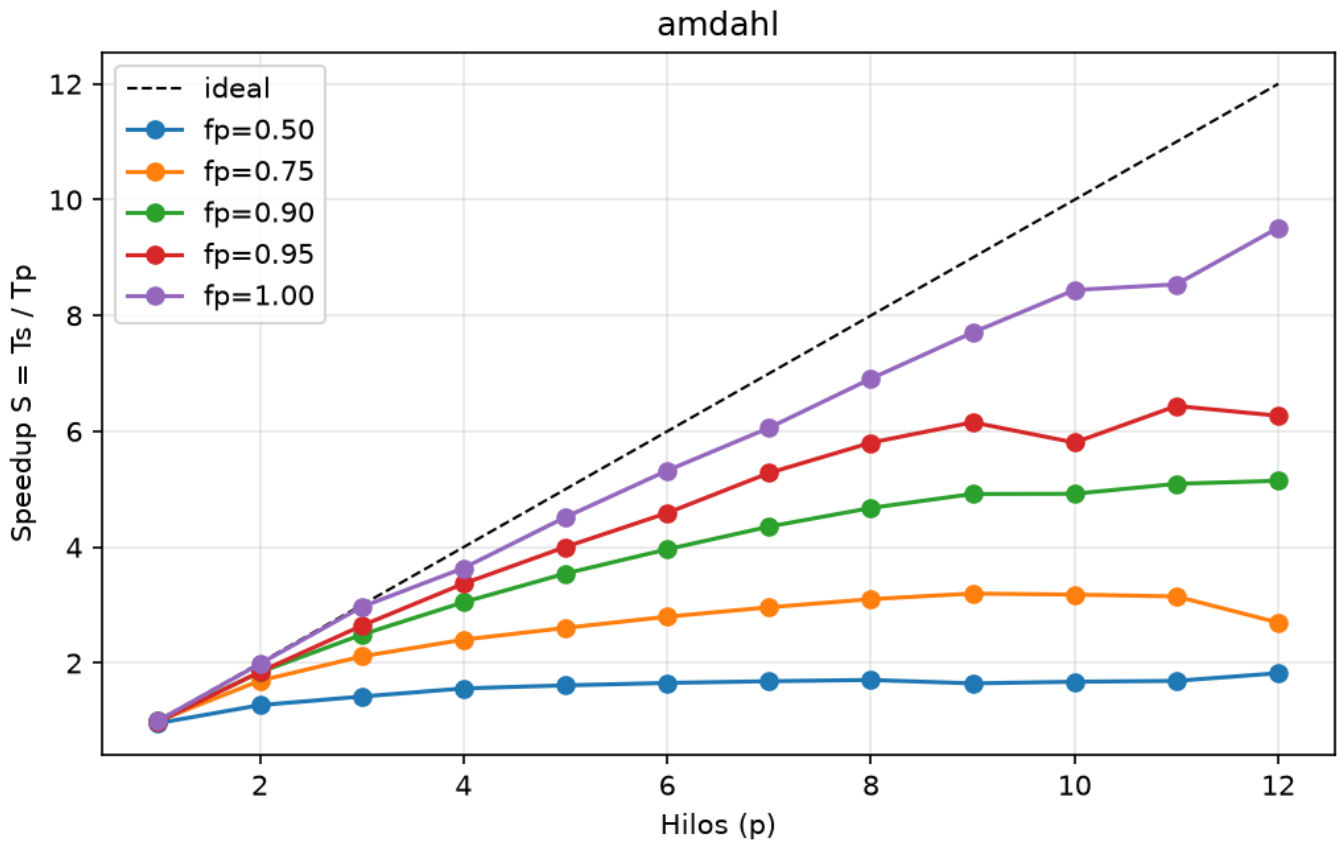
Ejecucion terminada. Codigo de salida: 0
La ventana permanecera abierta para la captura. Pulsa Enter para cerrarla:
```

Compilacion

```
> wsl.exe -d Ubuntu-24.04 -- make -B
gcc -O2 -Wall -fopenmp -o lab12 lab12_speedup.c

Ejecucion terminada. Codigo de salida: 0
La ventana permanecera abierta para la captura. Pulsa Enter para cerrarla:
```

Amdahl: speedup



Amdahl y suma

Amdahl

```
> wsl.exe -d Ubuntu-24.04 -- ./lab12 amdahl 400000 0.90 2
modo,N,p,param,t_seq,t_par,t_parte_seq
amdahl,400000,1,fp=0.90,0.656931,0.661172,0.066230
amdahl,400000,2,fp=0.90,0.656931,0.365744,0.067249
(checksum 9.719e+11)

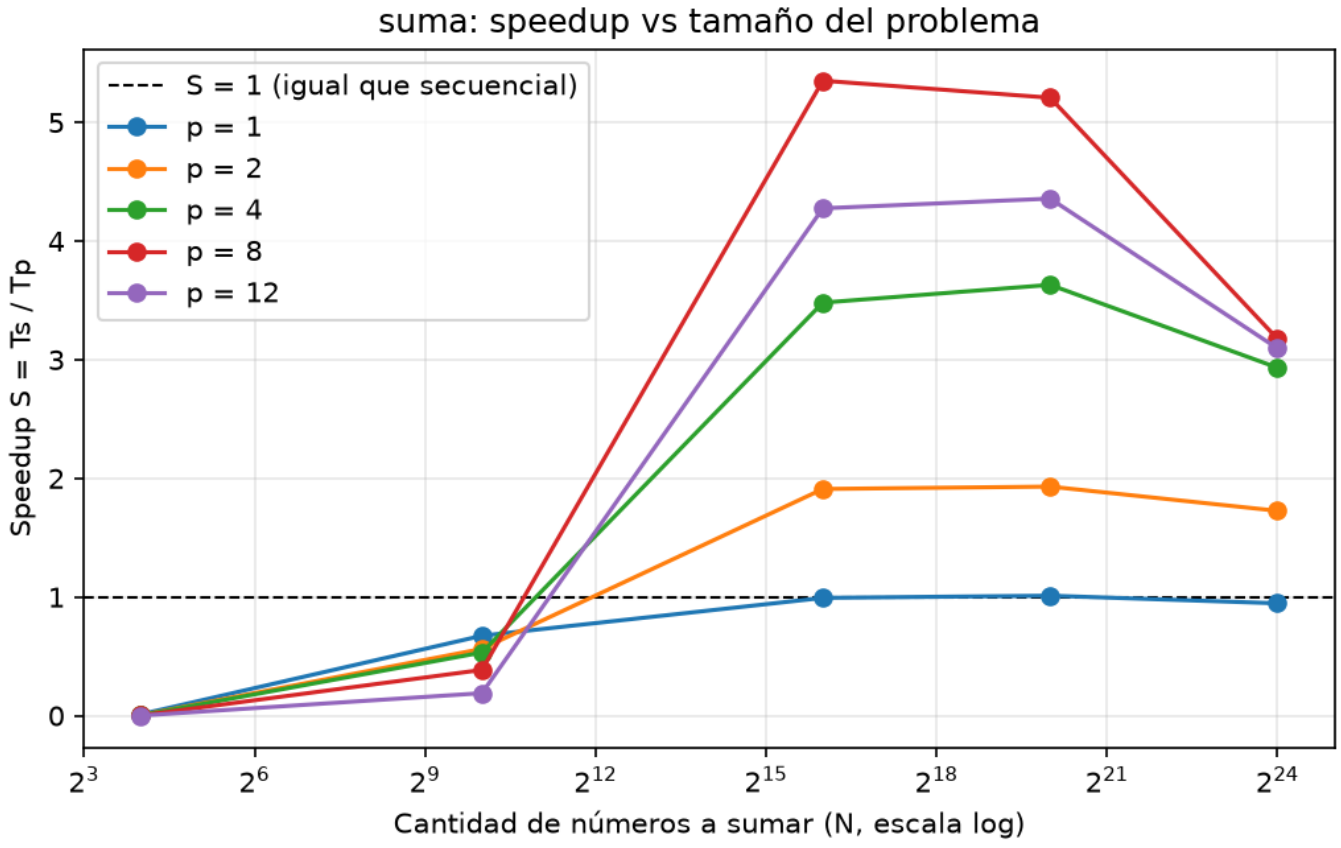
Ejecucion terminada. Codigo de salida: 0
La ventana permanecera abierta para la captura. Pulsa Enter para cerrarla:
```

Suma

```
> wsl.exe -d Ubuntu-24.04 -- ./lab12 suma 2
modo,N,p,param,t_seq,t_par,t_parte_seq
suma,16,1,R=100000,0.000000004,0.000000342,0
suma,16,2,R=100000,0.000000004,0.000000857,0
suma,1024,1,R=48828,0.000000773,0.000001132,0
suma,1024,2,R=48828,0.000000773,0.000001059,0
suma,65536,1,R=762,0.000052518,0.000052255,0
suma,65536,2,R=762,0.000052518,0.000027753,0
suma,1048576,1,R=47,0.000876271,0.000912559,0
suma,1048576,2,R=47,0.000876271,0.000442624,0
suma,16777216,1,R=3,0.014117731,0.013893552,0
suma,16777216,2,R=3,0.014117731,0.007273730,0
(checksum 1.072e+07)

Ejecucion terminada. Codigo de salida: 0
La ventana permanecera abierta para la captura. Pulsa Enter para cerrarla:
```

Suma: speedup



Desbalance y Gustafson

Desbalance

```
> wsl.exe -d Ubuntu-24.04 -- ./lab12 desbalance static 40000 2
modo,N,p,param,t_seq,t_par,t_parte_seq
desbalance,40000,1,static,0.065447,0.064597,0
desbalance,40000,2,static,0.065447,0.048310,0
(checksum 1.200e+10)

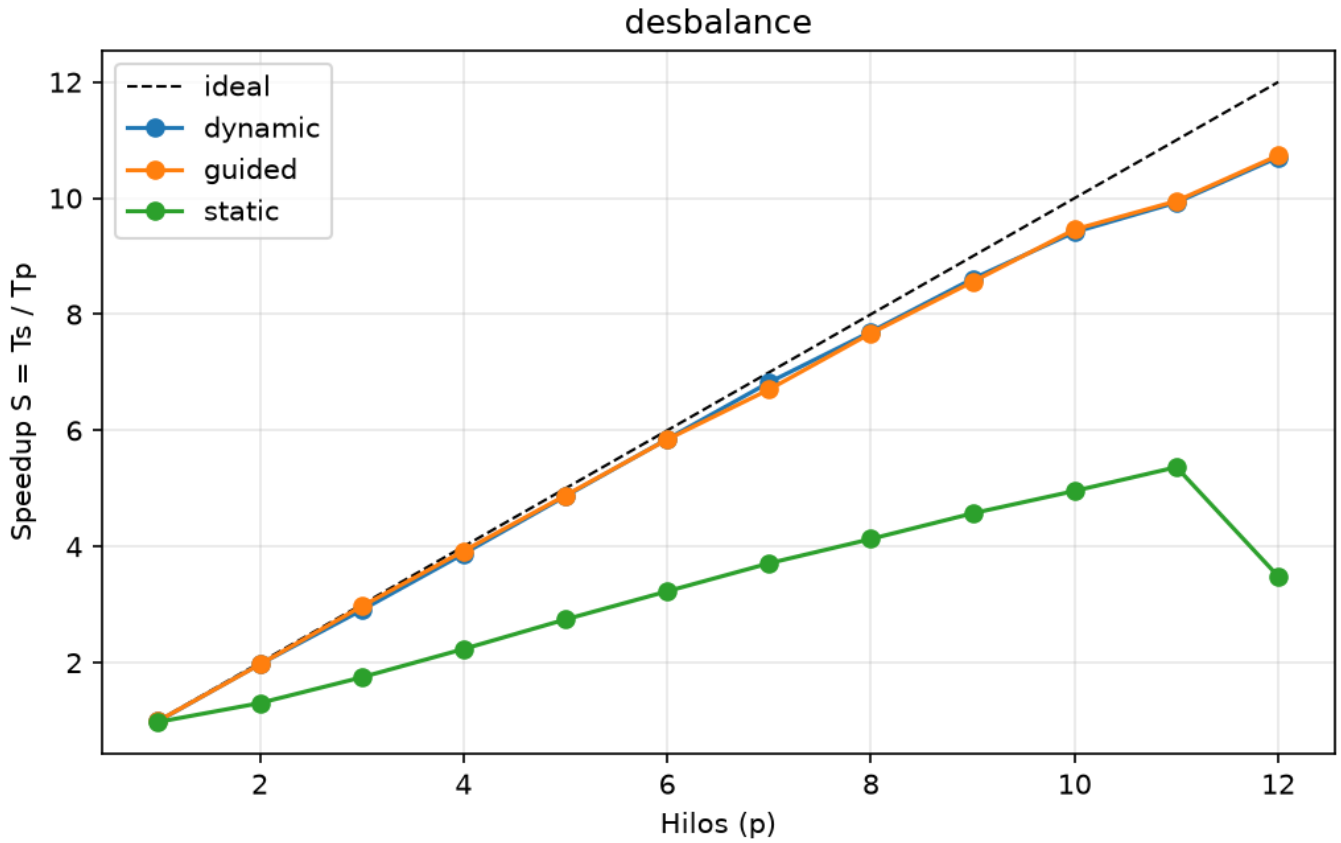
Ejecucion terminada. Codigo de salida: 0
La ventana permanecera abierta para la captura. Pulsa Enter para cerrarla:
```

Gustafson

```
> wsl.exe -d Ubuntu-24.04 -- ./lab12 gustafson 100000 20000 2
modo,N,p,param,t_seq,t_par,t_parte_seq
gustafson,120000,1,Wseq=20000,0.194377,0.196126,0.032375
gustafson,220000,2,Wseq=20000,0.349136,0.191516,0.031871
(checksum 2.500e+11)

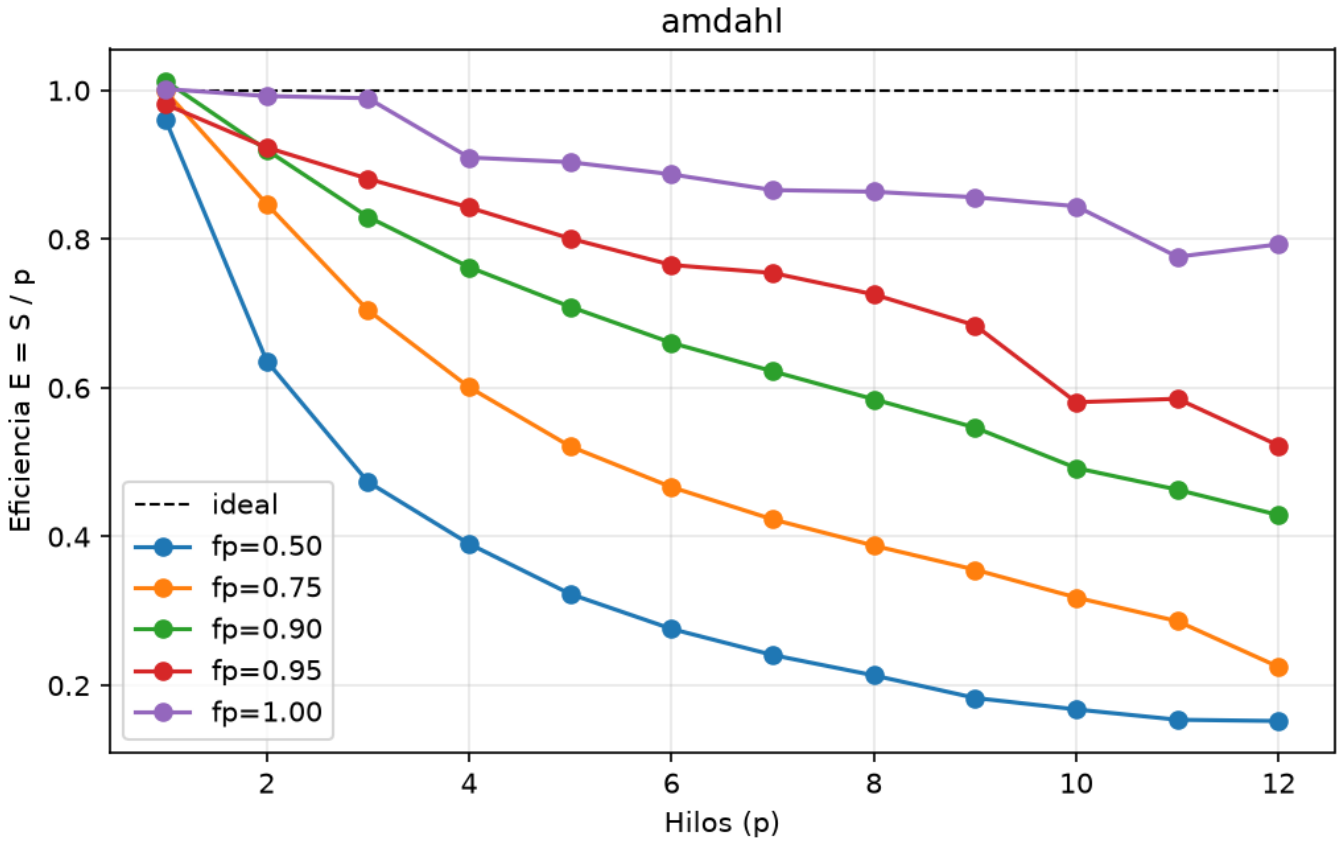
Ejecucion terminada. Codigo de salida: 0
La ventana permanecera abierta para la captura. Pulsa Enter para cerrarla:
```

Desbalance: speedup

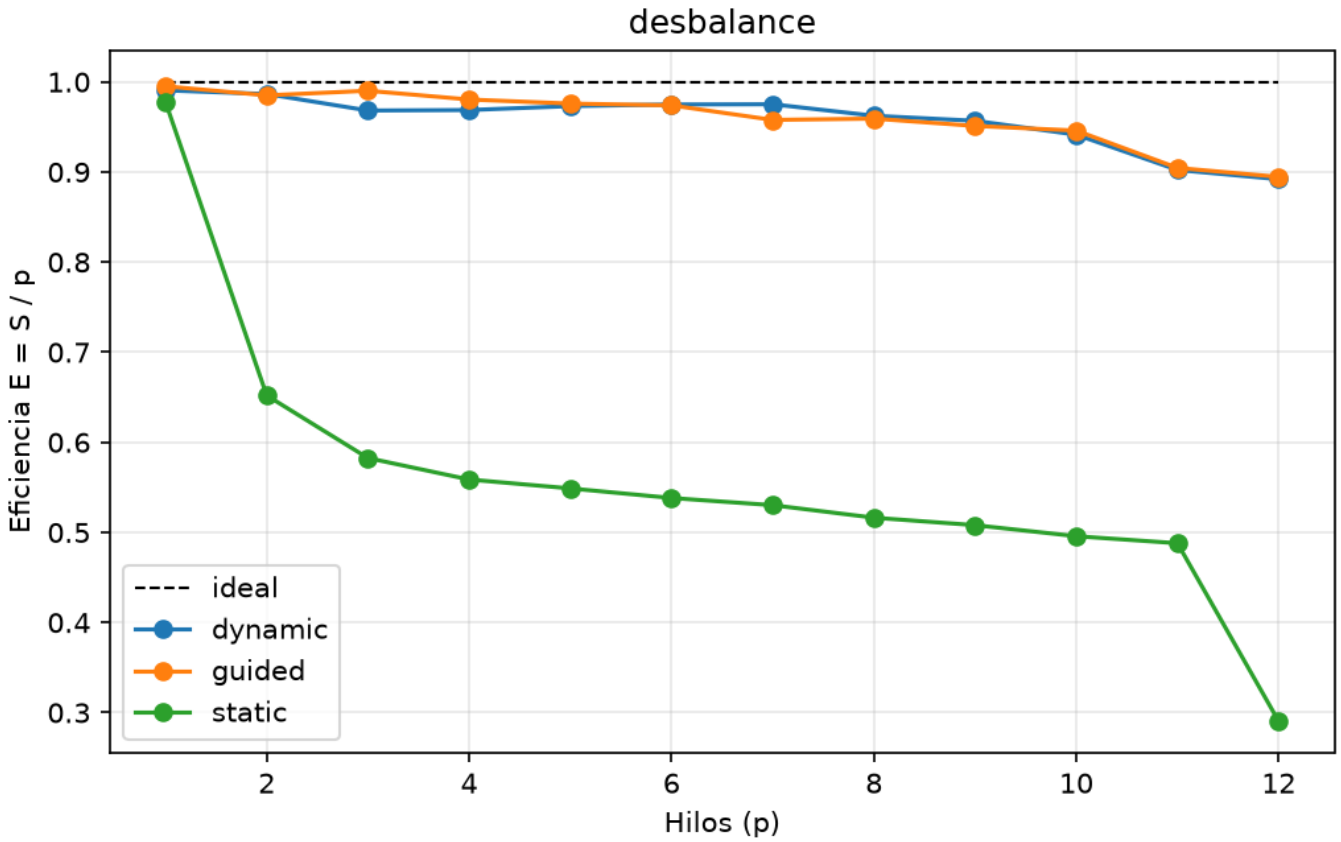


Eficiencia

Amdahl: eficiencia

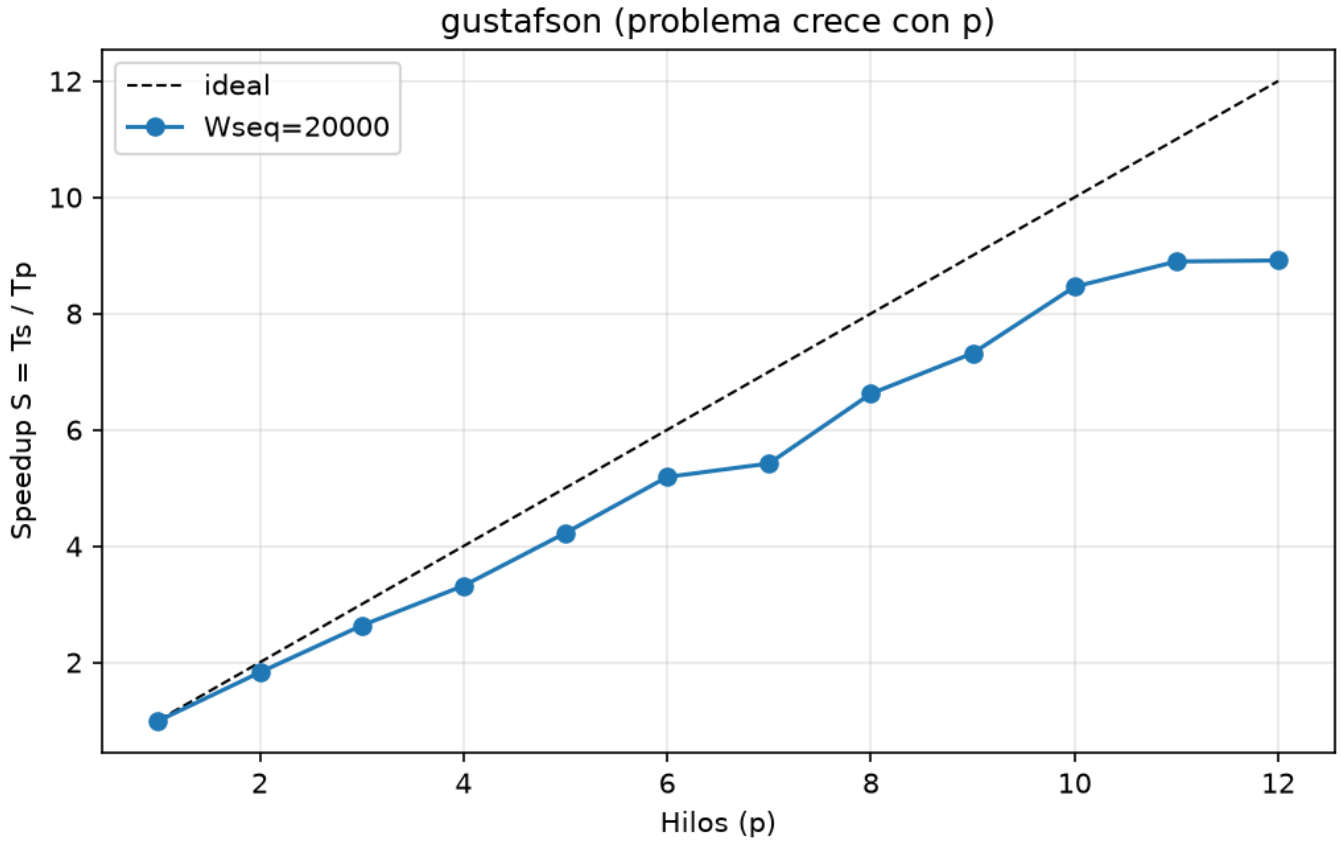


Desbalance: eficiencia



Gustafson

Gustafson: speedup



Gustafson: eficiencia

